Jesús M. Rueda-Becerril | Ph.D.

Astrophysicist – Data Scientist
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Profile

Data Scientist with research and industry experience, strong mathematical foundations, and expertise in Python, geospatial analysis, and machine learning. Skilled in designing scalable analytical tools, optimizing scientific code, and translating complex data into actionable insights. Experienced collaborating with cross-functional research teams to deliver production-quality scientific software.

Skills

Languages: Python, C/C++, Fortran 95, R, SQL, Shell, Rust, Java, Julia

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Geospatial & Remote Sensing: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

Simulation: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs (advection-diffusion, Fokker-Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Education

Ph.D. in Physics — **Oct. 2011 – Jul. 2017**
Universitat de València, Spain — Excellent *cum laude.*
Advisor: Prof. Miguel A. Aloy & Dr. Petar Mimica.
Thesis: *Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows*

M.Sc. in Physics — **Aug. 2009 – Sep. 2011**
Universidad Michoacana de San Nicolás de Hidalgo, Mexico
Advisor: Prof. José A. Cervera
Thesis: *Study of TOV stars with the SPH method*

B.Sc. in Physics — **Aug. 2004 – Dec. 2008**
Universidad Autónoma del Estado de México, Mexico — *"Dr. Juan Josafat Pichardo" Award*
Advisor: Prof. Francisco S. Guzmán
Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Experience

Independent Contractor — **Jan 2026 — Present**
Mercor
- Provide expert-level knowledge in Physics, Astrophysics, and Mathematics to improve models for a top AI research lab
- Assess model reasoning quality, identify failure modes, and provide structured feedback to improve scientific accuracy
- Apply understanding of mathematical foundations to benchmark model performance on complex STEM problems

Spatial Data Scientist — **May 2025 — Jan 2026**
TealWaters
- Lead project to optimize code for topographic terrain modeling and elevation derivative calculation.
- Engineered Python tools for geospatial data processing and analysis, utilizing QGIS for visualization and analysis.
- Utilize Wetland Intrinsic Potential (WIP) tool with random forest for wetland probability mapping in Skykomish watershed.
- Performed EDA, feature engineering, and geospatial data preparation, including multispectral (Sentinel 1/2) processing for ML land-cover and wetland probability models.
- Share knowledge and results with managers and decision-makers, train team members on WIP tool usage, and collaborate with software engineers and scientific team to transition prototypes into production.

Independent Research/Open-Source Developer — **Jan 2024 — May 2025**
- Developed scientific codes (Tleco, WindsOfChange), completed ML/AI coursework, and contributed to open-source geospatial and scientific-computing tools.

Software Engineer (Remote) **Apr 2022 – Jan 2024**

Paychex, Rochester NY, USA; based in Seattle, WA, USA

- Conducted data preparation, validation, and analysis in SQL from Oracle EBS datasets.
- Developed Java Kafka consumers for the streamlined transfer of large volumes of client data across databases. Developed high quality code using Java, Spring Boot, Kafka, PL/SQL, and deployed into production using Jenkins and OpenShift, following software development best practices.
- Collaborated with stakeholders, other software developers and engineers, and senior leadership to assess product needs and meet code standards for continuous integration model.
- Created Splunk dashboards and alerts for analysis of production data.
- Developed and deployed Python tests to ensure software quality and continuous integration.

Postdoctoral Research Associate **Feb 2021 – Apr 2022**

Rochester Institute of Technology, Rochester, NY, USA

- Led a team of specialists on a NSF-sponsored project to upgrade the C code `PatchworkMHD` to perform HPC simulations using state-of-the-art numerical techniques to model supermassive black hole binaries.
- Implemented a new feature (black hole spin) to `PatchworkMHD`, making more realistic binary black hole simulations without impacting runtime.
- Designed the experiments and evaluated state-of-the-art mathematic and numerical algorithms implemented in `PatchworkMHD` by running simulations at *Frontera* supercomputer (TACC, UT at Austin).
- Worked in a detail-oriented manner to successfully benchmark and identify performance optimization opportunities of the scientific code.
- Mentored and collaborated with a graduate student to apply the machine learning algorithm *gradient descent* to adjust the parameters of the open-source code, `Paramo`, to classify observations of blazars (extra-galactic objects) from Fermi-LAT telescope.
- Participated in a multi-institutional collaboration to study binary Neutron Star mergers through HPC simulations, resulting in 2 publications that provided critical breakthrough insights of the physics underlying these events.
- Published 3 co-authored papers and mentored graduate students (2 Ph.D.).

Postdoctoral Research Fellow **Oct 2018 – Nov 2020**

Purdue University, West Lafayette, IN, USA

- Developed the open-source code, `Paramo`, a numerical code in Fortran 95 optimized with OpenMP to perform radiative transfer simulations in relativistic astrophysics scenarios.
- Obtained and led a NASA grant to explain the origin and nature of radiation from active galaxies (blazars) using numerical and statistical models for objects observed with NASA Fermi-LAT space telescope. This research helped to unify our understanding of the two main types of blazars, identifying that important physical constraints applied to both objects.
- Developed Python tools to calculate the loss of energy due of high-energy particles due to interactions spectrum and evolution in the context of gamma-ray burst afterglows by developing sophisticated numerical integration, and OpenMP optimized features to `Paramo`.
- Collaborated with a group of multidisciplinary scientists to develop Python scripts for statistically modeling the COVID-19 outbreak in Mexico and helped create scientific infographics and blogposts for Spanish-speaking populations to reduce the spread of misinformation.
- Published 1 first-author and 1 co-authored paper and mentored three graduate students (1 M.S. and 2 Ph.D.).

Postdoctoral Research Fellow **Jan – Sep 2018**

Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Michoacan, Mexico

- Developed a Python script that would process images of spinning black holes simulations from the numerical code `GRTrans` to provide an SVM with training data that would later predict radio images of actual black holes.
- Developed an open-source data analysis and visualization tool in Python to provide any user with accessible tools to calculate radiative transfer phenomena (spectra and light-curves) in relativistic astrophysics.
- Organized a workshop to train graduate students in the use of the high-volume data storage tool HDF5.

Graduate Research Assistant **Oct 2011 – Jul 2017**

Universitat de València, Burjassot, Valencia, Spain

- Independently developed Shell and Python scripts to build pipelines to run simulation of the radiative transfer code `C-SPEV` and perform data processing of output datasets in HDFS format, ensuring data quality and integrity for downstream analysis and model fit with observations.
- Developed Python scripts to perform exploratory data analysis on datasets from NASA Fermi-LAT telescope and from the Very Large Baseline Array (VLBA) of the National Radio Astronomy Observatory (NRAO), and build non-linear regression models.
- Independently constructed models from `C-SPEV` simulations for curve fitting, pattern recognition, and prediction of data from NASA telescopes.
- Conducted multiple analyses to identify patterns in spectra and light-curves that allowed the quantification of magnetization of plasma in blazars.
- Conducted an analysis that identified the importance of including both cyclotron and synchrotron radiation from non-relativistic to ultra-relativistic charged particles in blazar simulations.
- Implemented sophisticated numerical tools and data handling to `C-SPEV` that could calculate both discrete and continuous spectra from particle distributions with arbitrary shape, without impacting simulation runtime.
- Published 2 first-author papers.

Publications

Articles

[9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ **976**, 182, (2024), arXiv:2405.17581.

[8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS **513**, 5766–5779, (2022), arXiv:2201.07790.

[7] Lopez-Armengol, F. G., Etienne, Z. B., [. . .], **Rueda-Becerril, J. M.**, [. . .] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D **106**, 083015, (2022), arXiv:2112.09817

[6] Murguia-Berthier, A., Noble, S., [. . .], **Rueda-Becerril, J. M.**, [. . .] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ **919**, 95, (2021), arXiv:2106.05356

[5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS **501**, 4092–4102, (2021), arXiv:2009.02273.

[4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS **496**, 974–986, (2020), arXiv:1910.14049.

[3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS **468**, 1169–1182, (2017), arXiv:1612.06383.

[2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS **438**, 1856–1869 (2014), arXiv:1310.5441.

[1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D **80**, 084023 (2009), arXiv:1009.1250.

Proceedings

5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), arXiv:2011.13797.

4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), arXiv:2011.13805.

3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) **233**, 159 (2014), arXiv:1502.07882.

2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. **61**, 02007 (2013), arXiv:1309.4612.

1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser **42**, 012001 (2013), arXiv:1211.1794.

MERGING WITH PREAMBLE PART

Research Grants

NASA Fermi Cycle-12 Guest Investigator Program **Grant #121077**
A simple model to understand the blazar sequence, PI: Giannios, D., Co-I: **Rueda-Becerril, J. M.** *2019*

Grants and Fellowships

Oct. 2018 – Nov. 2020: **Fellowship** from the Mexican Federal Government for international postdoctoral studies awarded by the National Council of Science and Technology (CONACyT).

Jan. – Sep. 2018: **Fellowship** from the Mexican Federal Government under the *Program for the Professional Development of Higher Education Institutions*, awarded by the Secretariat of Public Education.

Sep. 2014 – Aug. 2016: **Fellowship** from the Mexican Federal Government to study abroad awarded by the National Council of Science and Technology (CONACyT).

Oct. 2011 – Jun. 2014: **Fellowship** *Santiago Grisolía* awarded by the Council of Education, Research, Culture and Sport of the Valencian Comunity, Spain.

Sep. 2009 – Aug. 2011: **Fellowship** for MSc studies at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo, granted by the Mexican Council of Science and Technology (CONACyT).

Jun. – Aug. 2007: **Fellowship** for a temporary stay (3 months) in a national research center under the *XVII summer of scientific investigation program* awarded by the Mexican Academia of Science.

Projects

Scientific ML (Python, scikit-learn, PyTorch) **Jan 2026 — Present**
Blazar Population ML *GitHub*

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data; inverse problem methodology directly analogous to ice dynamics parameter estimation from remote sensing observables.
- Developing neural surrogate model (Tleco emulator) to replace expensive Fokker-Planck numerical integration across a high-dimensional parameter space, enabling fast inference over physically motivated priors.

Open-Source Lagrangian Particle Simulation Library (R/C++) **Aug 2024 — Present**
WindsOfChange *GitHub*

- Developed C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Open-Source Physics Simulation Library **Jan 2024 — Sep 2024**
Tleco *GitHub*

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Open-Source Physics Simulation Code (CONAHCYT Funded) **Oct 2018 — Apr 2022**
Paramo *GitHub*

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

Large-Scale MHD Simulation Code (NSF Project) **Feb 2021 — Apr 2022**
PatchworkMHD

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

Awards

Marcos Moshinsky Award: for *Best Poster* presented at the IWARA 2020 Video Conference, Mexico City, 6 – 12 September 2020.

Lic. Juan Josafat Pichardo Cruz Award: for finishing and defending a licentiate thesis within a year after completing the undergraduate credits, granted by the Universidad Autónoma del Estado de México, 2009.

Invited Talks

The Role of Machine Learning in Environmental Science
Seminar on Climate Change Ecology, University of Washington, Seattle, WA, USA, March, 2024

Morphology of the spectra from numerical simulations of the internal shocks model for blazars
Astrophysics Seminar, Purdue University, West Lafayette, IN, USA, February 4, 2019

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars
DATA group weakly Seminar, Instituto de Astronomía, UNAM, Mexico City, Mexico, June 19, 2018

Numerical treatment of non-thermal radiation in the internal shocks model for blazars
Weekly Seminar, Instituto de Física y Matemáticas, Morelia, Mexico, March 2, 2018

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars
IVICFA's Fridays: Computation in Physics, IFIC, Paterna, Spain, October 17, 2014

Meetings and conferences

Contributed Talks

Simulations of supermassive binary black holes accretion dynamics in the spinning case
APS April Meeting 2022, New York City, NY, April 9–12, 2022

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows
APS April Meeting 2021, April 17–20, 2021

The blazar sequence revised
9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020
https://www.youtube.com/watch?v=BAZNWLNT69M

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario
Extreme-Astrophysics in an Ever-Changing Universe, Ierápetra, Greece, June 16–20, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets
XXXIV Biennial meeting of the Royal Spanish Society of Physics, Valencia, Spain, July 15–19, 2013

Poster Sessions

A numerical approach for radiative cooling in relativistic outflows
9th International Workshop on Astronomy and Relativistic Astrophysics, Video Conference, September 6–12, 2020
Marcos Moshinsky Award for Best Poster. https://www.youtube.com/watch?v=0TJiKg7kOPI

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars
Swift: 10 years of Discovery, Rome, Italy, December 2–5, 2014

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets
The Innermost Regions of Relativistic Jets and Their Magnetic Fields, Granada, Spain, June 10–14, 2013

Teaching & Mentoring Experience

Zachary Davis [8, 9] — **Mentoring**
Graduate student, Department of Physics and Astronomy, Purdue University — *2018 – 2022*

Amanda O. Harrison [5] — **Mentoring**
Graduate student, Department of Physics and Astronomy, Purdue University — *2018 – 2020*

Hao Zhang [4] — **Mentoring**
Graduate student, Department of Physics and Astronomy, Purdue University — *2018 – 2019*

Thermodynamics (Graduate Level) — **Guest Lecturer**
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo — *Jun 2018*

Professional Development

High Performance Computing on Frontera
Jason Allison et al., TACC, Austin, TX, USA, May 20, 27 and June 3, 2021 — *Lecture*

Writing Winning Grants
Dr. Lauren Broyles, Purdue University, West Lafayette, IN, USA, November 7, 2019 — *Lecture*

XSEDE HPC Workshop: Summer Boot Camp
John Urbanic, Purdue University, West Lafayette, IN, USA, June 3 – 6, 2019 — *Workshop*

Data Analysis and Machine Learning with Python
Dr. Alejandro Torres, Universitat de València, Burjassot, Spain, February 7 – 16, 2017 — *Workshop*

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit
Dr. Vassilios Mewes, Universitat de València, Burjassot, Spain, July 6 – 7, 2016 — *Workshop*
No. of hours: 8

The Universe in the Light of PLANCK and BICEP2
Prof. Nick Mavromatos, Universitat de València, Burjassot, Spain, May 23 – 16, 2014 — *Lecture series*
No. of credits: 2

Dark Matter
Prof. Alejandro Ibarra, Universitat de València, Burjassot, Spain, September 23 – 27, 2013 — *Lecture series*
No. of credits: 2

International Cagèse School on Cosmic Accelerators
Institut d'Études Scientifques de Cargèse, Cargèse, France, April 23 – May 8, 2013 — *Summer school*

Introduction to C++ Programming
Dr. Jacek Generowicz, Universitat de València, Burjassot, Spain, April 9 – 12, 2012 — *Workshop*
No. of credits: 6

Numerical Relativistic Astrophysics
Prof. Luciano Rezzolla, Universitat de València, Burjassot, Spain, March 27 – April 4, 2012 — *Lecture series*
No. of hours: 9

Fortran for Scientific Computing
HLRS, University of Stuttgart, Stuttgart, Germany, Mar. 5 – 9, 2012 — *Workshop*
No. of hours: 33

Certifications and Credentials

Gen AI Intensive — **Kaggle**
Generative AI

Data Scientist Professional with Python — **DataCamp**
Python Programming, Data Science, Data Communication, Machine Learning

Machine Learning Scientist with Python — **DataCamp**
Machine Learning, NLP, Deep Learning, Image Processing, Big Data

Mathematical Foundations of Machine Learning — **Udemy**
Python Programming, Data Science, Statistics, Machine Learning

Python for Statistical Analysis — **Udemy**
Python Programming, Data Science, Data Communication, Machine Learning

Outreach

Los más rápidos y los más furiosos *(The Fastest and the Most Furious)* — **Online talk**
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco — *September 4, 2020*
Tabasco, Mexico

Una simulación de la física y la astrofísica *(A Simulation of Physics and Astrophysics)* — **Online talk**
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco — *August 14, 2020*
Tabasco, Mexico

ANITA y la teoría de los universos paralelos *(ANITA and the Theory of Parallel Universes)* — **Blog post**
Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post /anita — *May 29, 2020*

¿Qué es el modelo SIR? *(What is the SIR Model?)* — **Blog post**
Científicos Mexicanos en el Extranjero, mexiciencia.github.io /post/modelo-sir — *May 25, 2020*

Evolución del brote epidémico de COVID-19 *(Evolution of the COVID-19 Epidemic Outbreak?)* — **Blog post**
Científicos Mexicanos en el Extranjero, mexiciencia.github.io/post /covid19 — *April 5, 2020*
Collaborator with the data analysis/modeling

Annual Department of Physics and Astronomy Poster Event — **Posters (3) presentation**
Department of Physics and Astronomy, Purdue University — *November 13, 2019*
West Lafayette, IN, USA

Post-Doc Panel Q&A: What Happens When we Complete our PhDs? — **Panelist**
Department of Physics and Astronomy, Purdue University — *April 10, 2019*
West Lafayette, IN, USA

Annual Department of Physics and Astronomy Poster Event — **Poster presentation**
Department of Physics and Astronomy, Purdue University — *November 14, 2018*
West Lafayette, IN, USA

¿Decía Einstein la verdad? *(Was Einstein Telling the Truth?)* — **Talk**
Facultad de Ciencias, Universidad Autónoma del Estado de México — *March 11, 2009*
Toluca, Mexico

Synergetic Activities

X Scientific Meeting of the Spanish Astronomical Society — **Valencia, Spain**
Organizing contributor, 14–16 December, 2012

Other activities

Aug 2007– May 2009: *Representative* of the Physics students community at the Governing Council of the Faculty of Sciences of the Universidad Autónoma del Estado de México.

Languages

Spanish: Native proficiency

English: Full professional proficiency

Catalan: Intermediate